

System **Architecture**

Fundamentals

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Objective



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- ❑ **Describe** the fundamental concepts and guiding principles of system architecture.
- ❑ **Distinguish** between logical architecture and physical architecture in terms of structure, purpose, and representation.
- ❑ **Evaluate and compare** centralized, distributed, and hybrid architectural models based on their characteristics and applications.
- ❑ **Assess** the suitability of different architectural approaches according to specific organizational requirements and operational contexts.

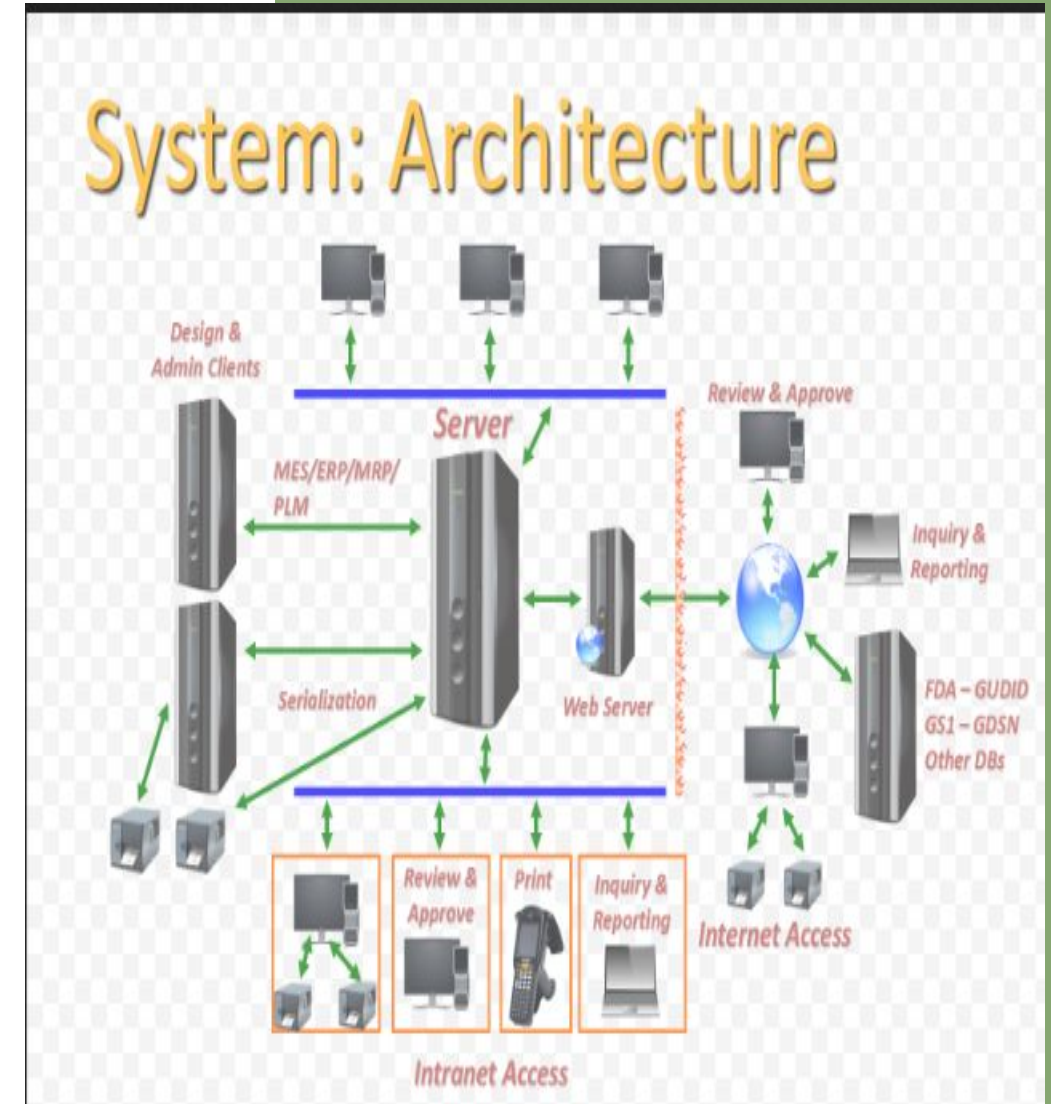
I. Architectural Concepts and Principles:

What is System Architecture?

System Architecture is the conceptual design that defines:

- ❑ Structure of a system
- ❑ Components and modules
- ❑ Relationships between components
- ❑ Rules governing system interaction

It serves as the **blueprint** of an information system.



Core Architectural Principles



Modularity - System divided into independent modules.



Scalability - Ability to handle growth



Reliability - System performs consistently



Maintainability - Easy to modify and update

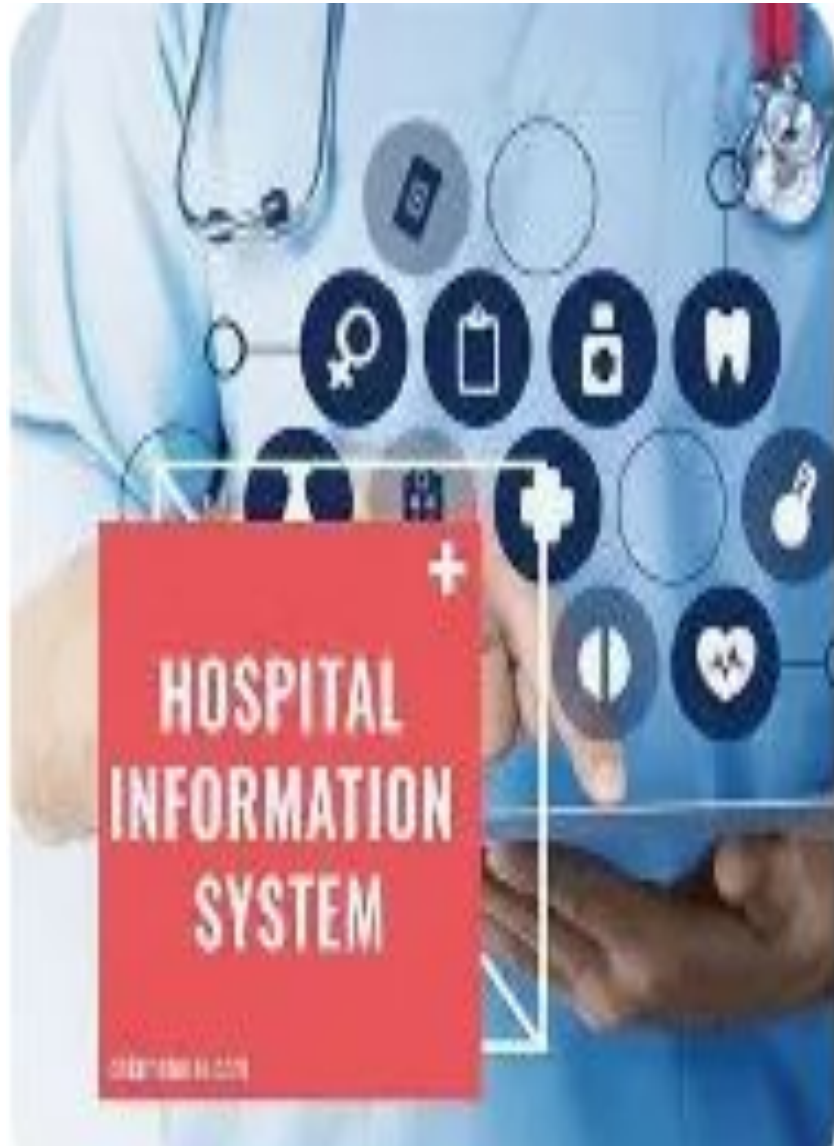


Security - Protection against unauthorized access



Interoperability - Ability to work with other system

Ex: Hospital Information System (HIS)



A **Hospital Information System (HIS)** manages:

- ❑ Patient registration
- ❑ Electronic medical records (EMR)
- ❑ Laboratory results
- ❑ Billing and insurance
- ❑ Pharmacy management

Using this System: give me a sample regarding the architectural principle applied to the system:

- ❑ Modularity
- ❑ Scalability
- ❑ Reliability
- ❑ Maintainability
- ❑ Security
- ❑ Interoperability

II. Logical vs Physical Architecture



Logical Architecture

A high-level conceptual representation of system functions and relationships. It answers the question **“How does the system work?”**

Focuses on:

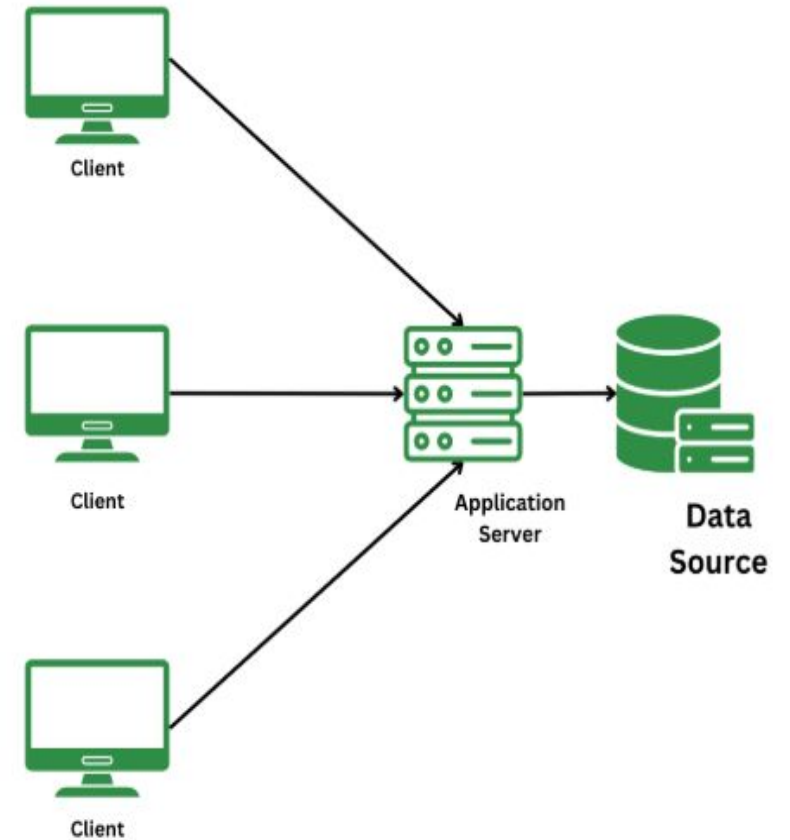
- What the system does
- Functional components
- Data flow

Note: Does NOT show hardware details

Example:

- Web Application
- Database
- Authentication Service

Three Tier Architecture



II. Logical vs Physical Architecture



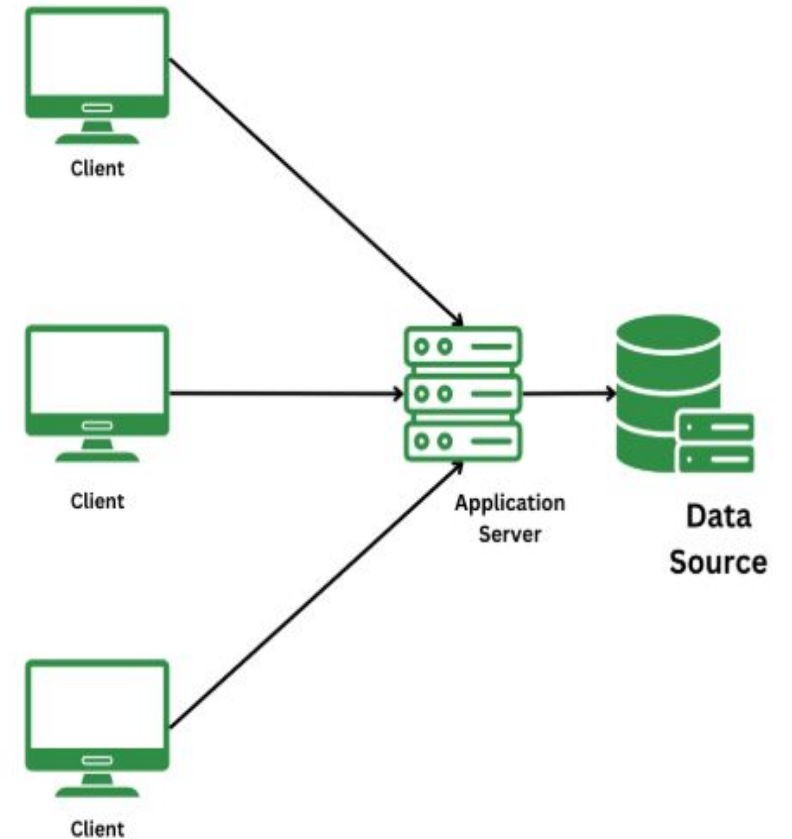
Logical Architecture

Typical Logical Layers:

- 1 **Presentation Layer**
 - ❖ User interface
 - ❖ Web or mobile screens
- 2 **Application Layer**
 - ❖ Business logic
 - ❖ Processing rules
- 3 **Data Layer**
 - ❖ Database
 - ❖ Data storage and retrieval

This diagram shows functions and relationships — not actual servers.

Three Tier Architecture



II. Logical vs Physical Architecture



Physical Architecture

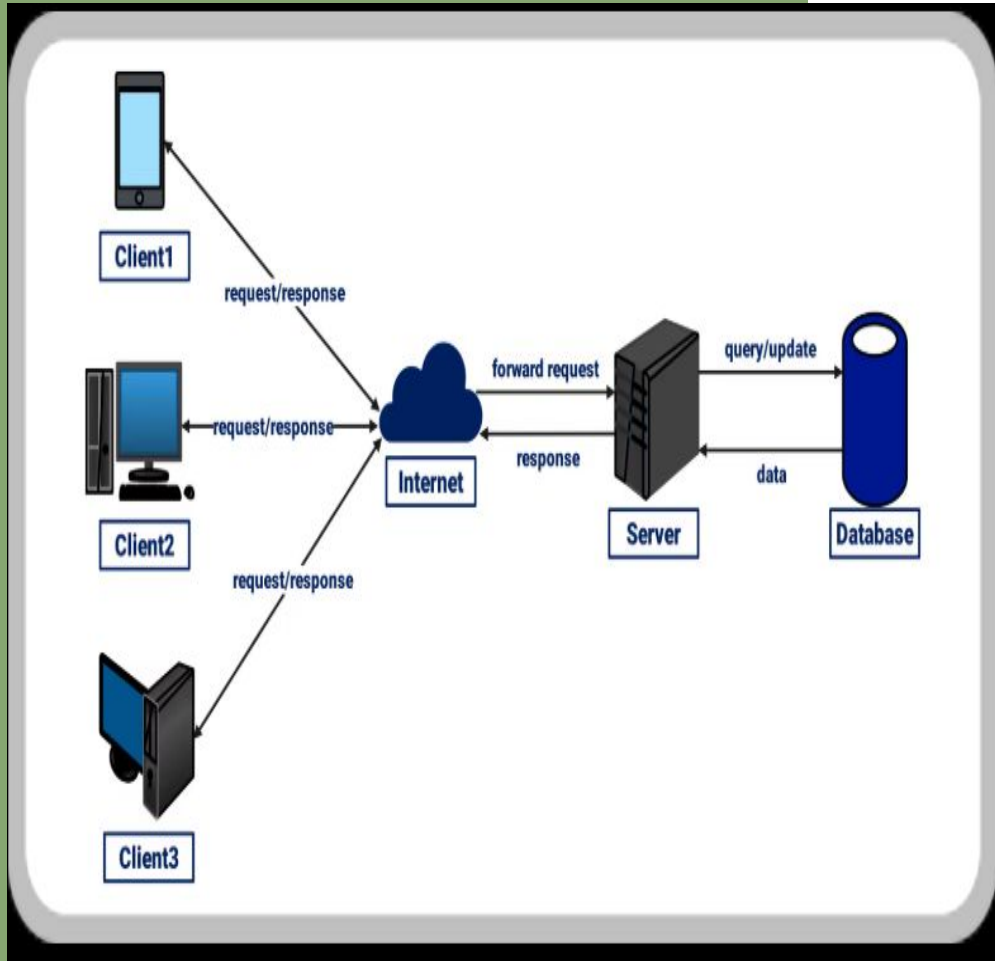
Physical architecture is the detailed representation of the actual hardware, infrastructure, and network components where a system is deployed and operated

It shows:

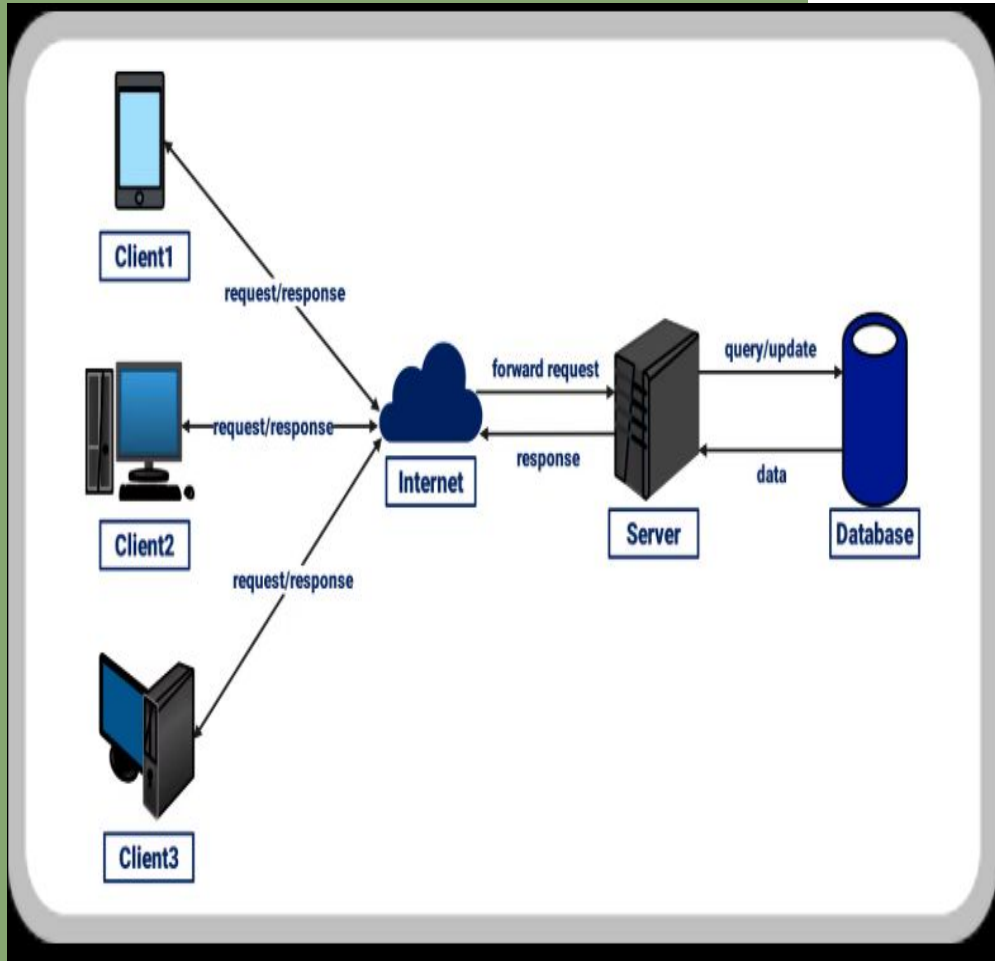
- ❖ Servers
- ❖ Computers
- ❖ Network devices
- ❖ Data centers
- ❖ Cloud infrastructure
- ❖ Storage systems

It answers the question:

“Where and on what does the system run?”



II. Logical vs Physical Architecture



Example (Hospital Information System)



Logical Architecture shows:

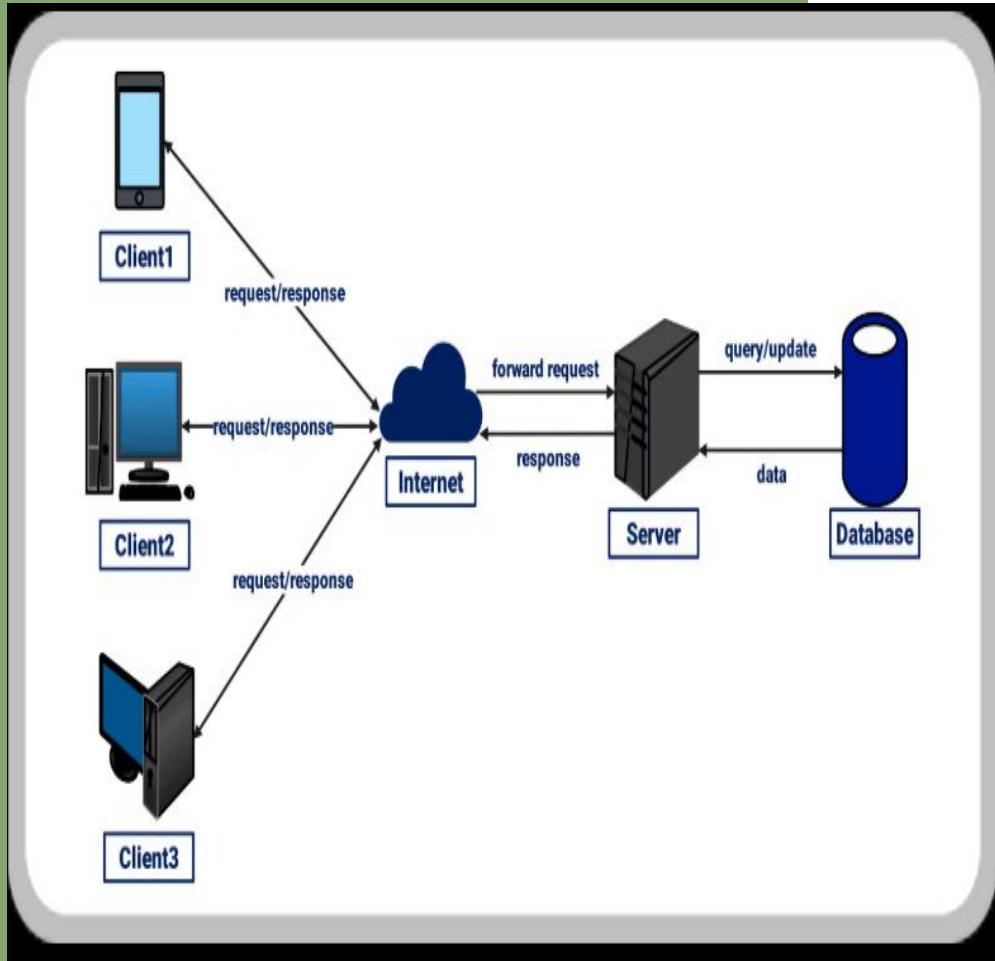
- ❖ EMR module
- ❖ Billing module
- ❖ Lab module



Physical Architecture shows:

- ❖ EMR hosted on Server 1
- ❖ Database on Server 2
- ❖ Backup server in another building
- ❖ Secure VPN for remote clinics

II. Logical vs Physical Architecture



Why Physical Architecture Is Important

- ✓ Ensures system performance
- ✓ Supports scalability
- ✓ Enhances security planning
- ✓ Enables disaster recovery
- ✓ Prevents single points of failure

III. Centralized, Distributed, and Hybrid Architecture

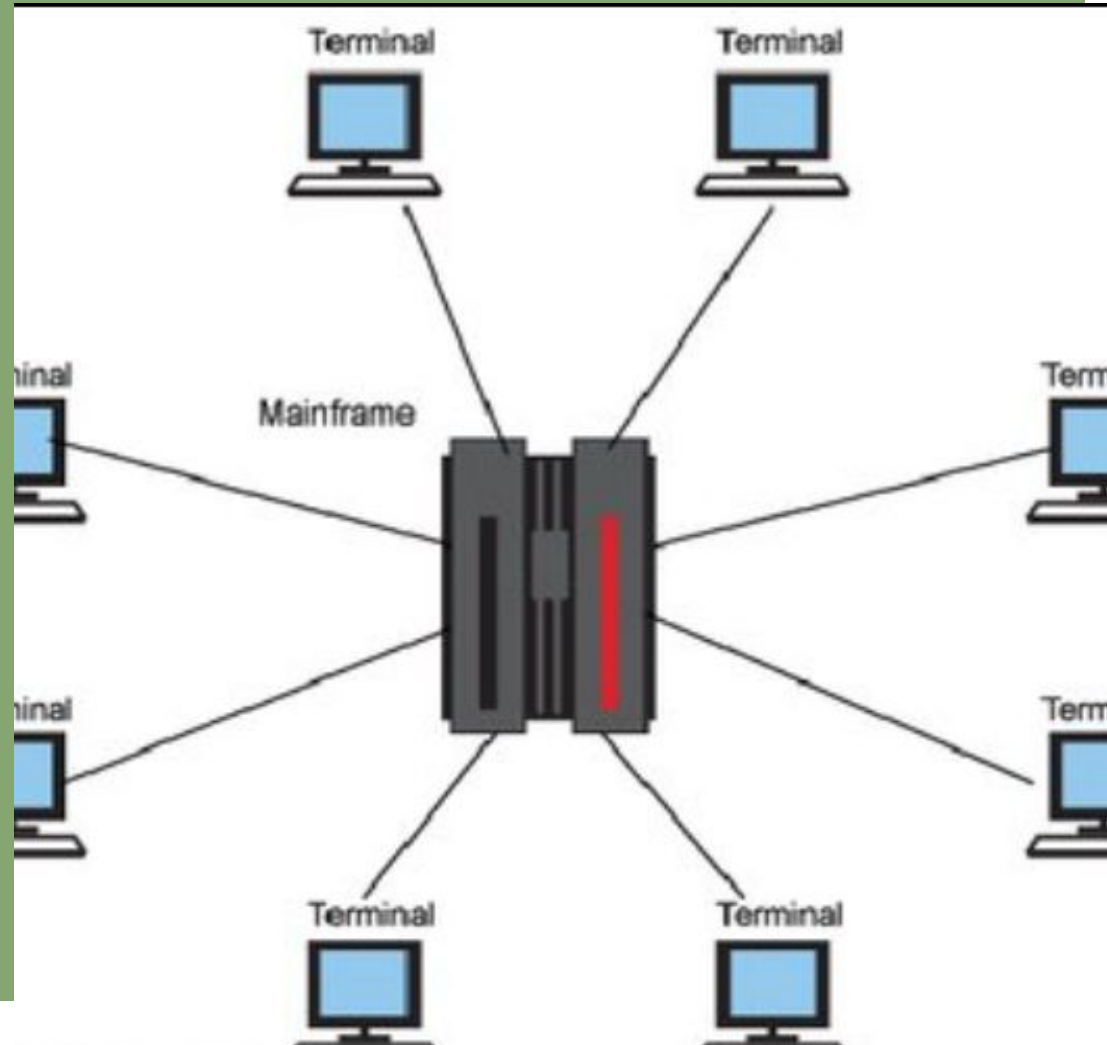
① Centralized Architecture

Centralized architecture is a system design where all major processing, data storage, and control are handled by a single central server or main system.

All users or client devices connect to this central system to access services and data

It answers the question:

“Where does the processing happen?” →
In one central location.



III. Centralized, Distributed, and Hybrid Architecture

① Centralized Architecture

All processing occurs in a single central server

Characteristics

- ❖ Central database
- ❖ Central control
- ❖ Easier management

Advantages

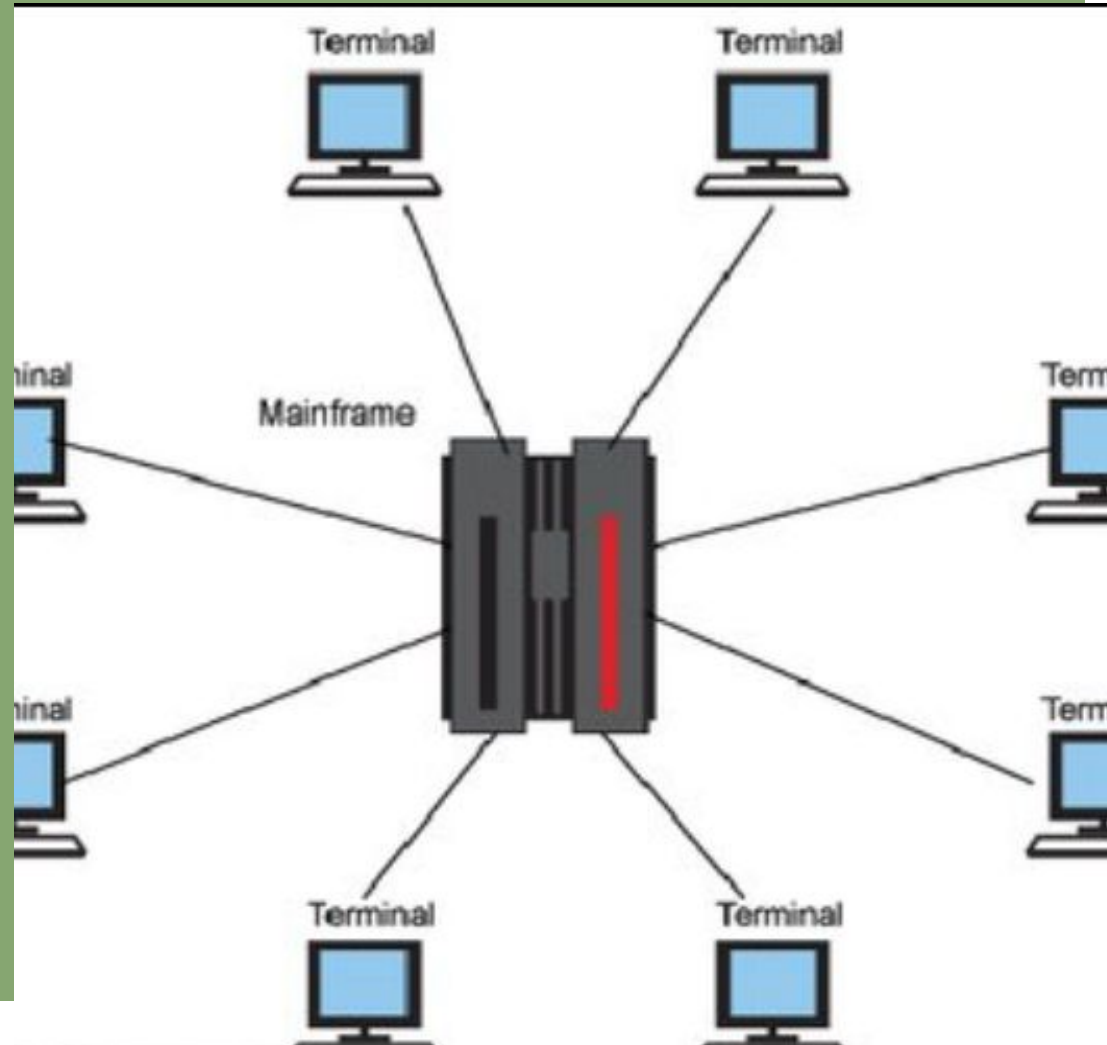
- ❖ Strong security control
- ❖ Easier maintenance
- ❖ Simplified backup

Disadvantages

- ❖ Single point of failure
- ❖ Performance bottlenecks
- ❖ Limited scalability

Example

Mainframe-based systems.



III. Centralized, Distributed, and Hybrid Architecture

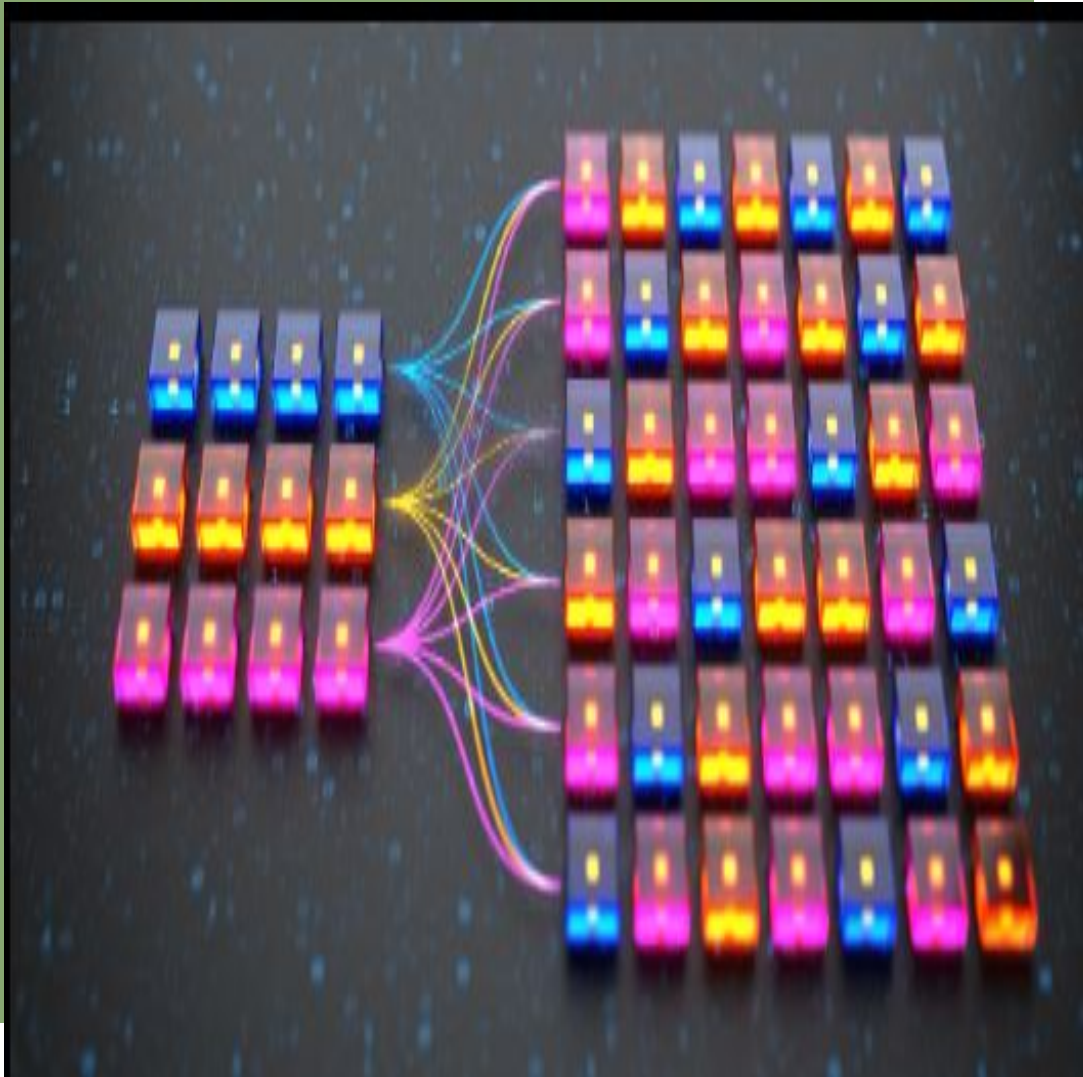
② Distributed Architecture

Distributed architecture is a system design where processing, data storage, and services are spread across multiple servers or locations instead of being handled by a single central system.

It answers the question:

“Where does processing happen?”

→ **In multiple locations.**



III. Centralized, Distributed, and Hybrid Architecture

② Distributed Architecture

Processing is distributed across multiple systems or locations.

Characteristics

- ❖ Multiple servers
- ❖ Decentralized processing
- ❖ Network-based coordination

Advantages

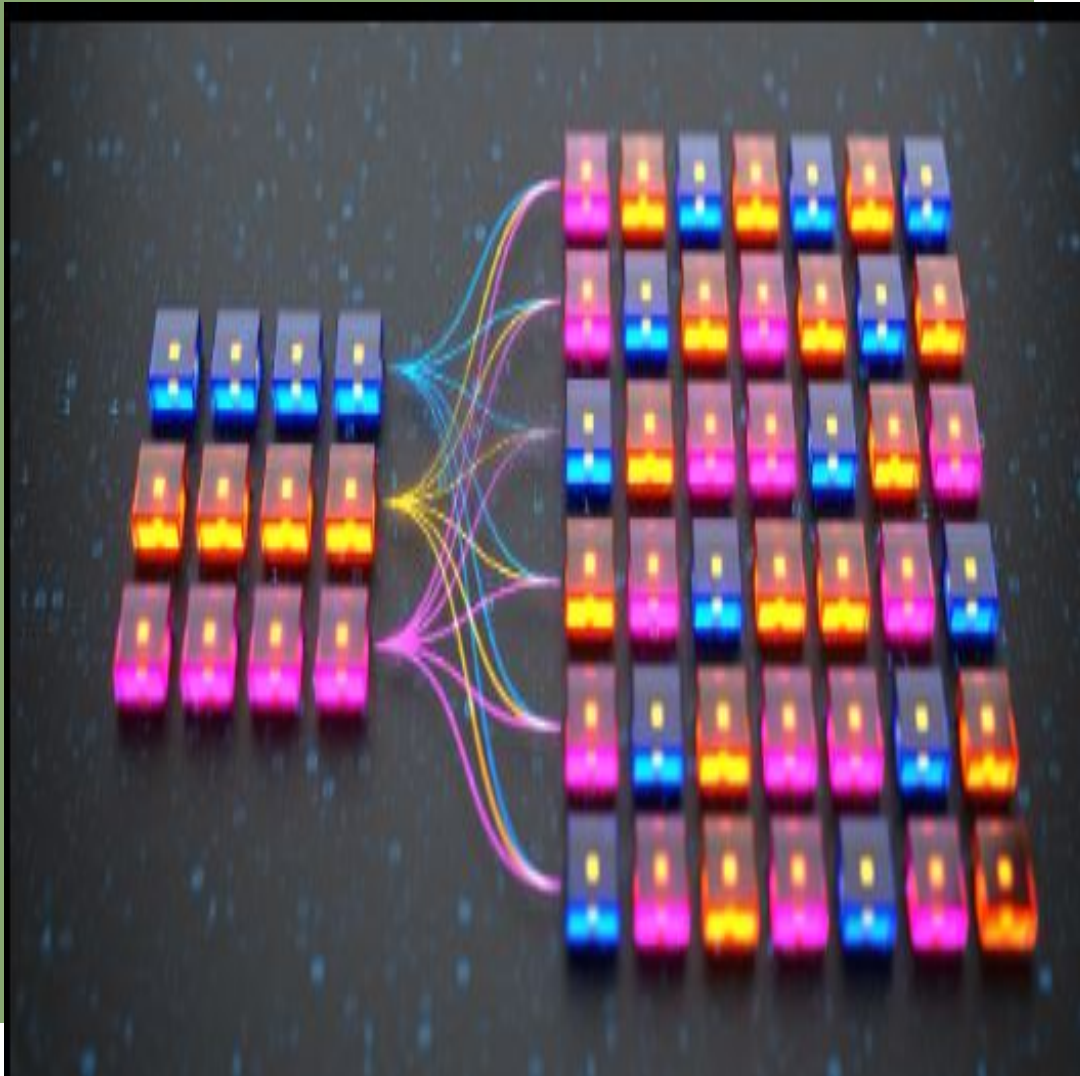
- ❖ High scalability
- ❖ Fault tolerance
- ❖ Better performance

Disadvantages

- ❖ Complex management
- ❖ Network dependency
- ❖ Security challenges

Example

Cloud computing, microservices architecture.



III. Centralized, Distributed, and Hybrid Architecture

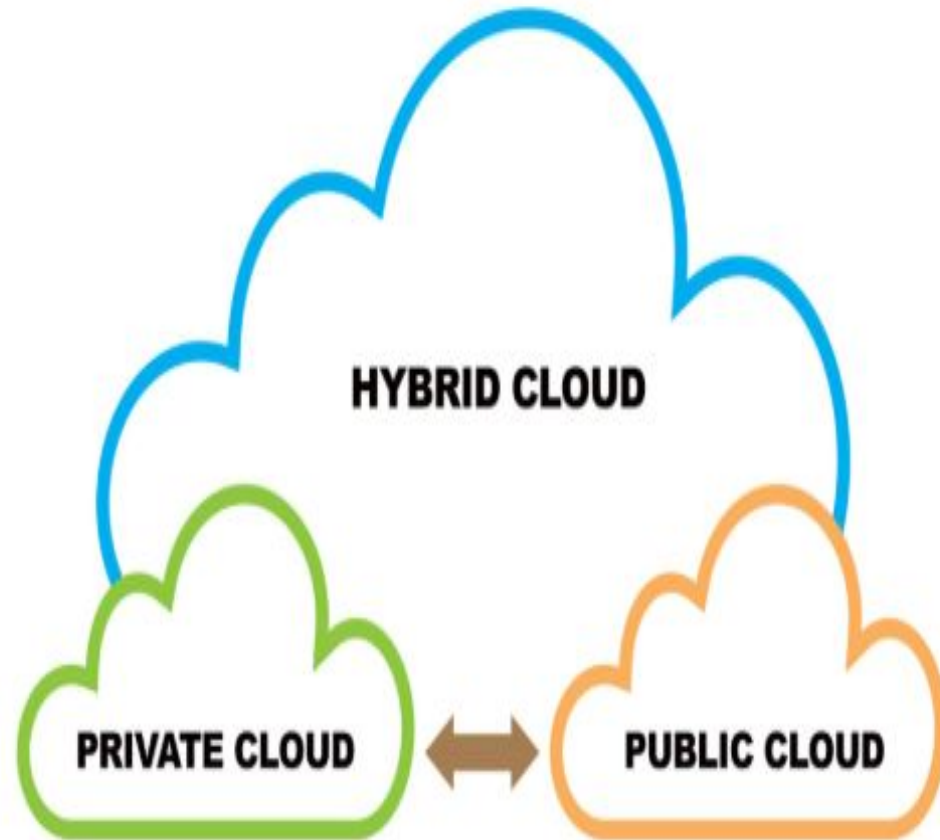
③ Hybrid Architecture

Hybrid architecture is a system design that combines elements of both centralized and distributed architectures.

Some components are centrally managed, while others are distributed across multiple servers or locations

It answers the question:

“Can we balance central control and distributed flexibility?” → Yes, through hybrid architecture.



III. Centralized, Distributed, and Hybrid Architecture

③ Hybrid Architecture

Combination of centralized and distributed models.

Characteristics

- ❖ Core systems centralized
- ❖ Certain services distributed

Advantages

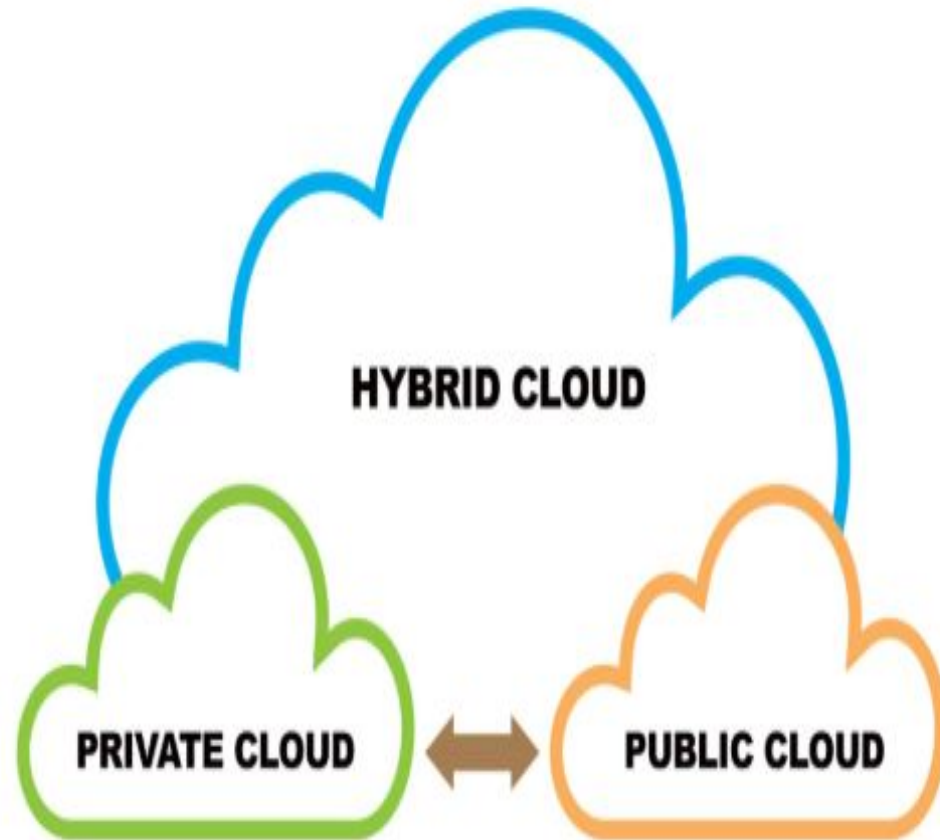
- ❖ Balanced control and flexibility
- ❖ Optimized performance
- ❖ Gradual modernization

Disadvantages

- ❖ Architectural complexity
- ❖ Requires strong governance

Example

- ❖ Central database
- ❖ Regional application servers
- ❖ Cloud-hosted services



Class Activity

Activity:

Identify one system and determine if its centralized, distributed or hybrid (except: Student information system and hospital info system)

List the physical and logical architecture of the system

Thank You

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